

**A LONGITUDINAL STUDY OF THE INFLUENCE OF ALLIANCE NETWORK
STRUCTURE AND COMPOSITION ON FIRM EXPLORATORY INNOVATION**

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Abstract

This study examines the influence of the structure and composition of a firm's alliance network on its *exploratory innovation*. In a longitudinal investigation of 77 telecommunications equipment manufacturers, I find the technological diversity of a firm's alliance partners increases its exploratory innovation. I also find that network density among a firm's alliance partners strengthens the influence of diversity. These results suggest the benefits of network closure and access to diverse information can coexist in a firm's alliance network and the combination of the two increases exploratory innovation.

A core area of research on strategic alliances examines their influence on firm performance (Gulati, 1998). Within this domain of inquiry, research often characterizes alliances as wellsprings of innovation and new capabilities (e.g., Badaracco, 1991; Hamel, 1991). Many studies show the alliance networks in which firms are embedded can enhance firm learning and innovation (e.g., Ahuja, 2000; Shan, Walker, & Kogut, 1994; Smith-Doerr et al., 1999; Soh, 2003). Despite this evidence, substantial opportunity exists to expand our understanding of how and under what conditions alliance networks influence firm innovation. A systematic review of nearly 40 years of research published in 17 leading management and disciplinary social science journals (Phelps & Heidl, 2008) revealed the existing literature on alliances and firm innovation is limited in at least five important respects: (1) an emphasis on network structure and alliance characteristics with little attention to network composition, in general, and to network-level technological diversity, in particular, (2) conflicting results about the impact of network structure on firm innovation, (3) a widely held, yet largely unexamined, assumption that a firm's access to diverse information and the innovation benefits of network closure are mutually exclusive, (4) an inattention to the potential complementarities between network structure and composition, and (5) an emphasis on the volume of firm innovation, with little consideration of its novelty or exploratory content.

While some research has examined the influence of alliance network structure on firm innovation, the resources and other attributes of a firm's partners (i.e., "network composition") have received little attention. Indeed, recent research has recognized alliance network studies have, in general, largely ignored network composition and has called for greater attention to partner attributes in network research (Gulati, 2007; Lavie, 2006; Maurer & Ebers, 2006). The few studies that have examined both structure and composition have focused on the strength of partner technological resources as the measure of composition (e.g., Baum et al., 2000; Stuart, 2000). This research shows the innovativeness of a firm's partners improves its innovation performance, independent of the number of its alliances. However, research has not yet explored the influence

of the technological diversity that inheres in a firm's alliance network¹. Such compositional diversity is particularly relevant to a current debate in the social network and alliance literatures.

Research that examines the influence of social networks on creativity and innovation stresses the benefits actors derive from network structure and explores how these benefits, or "structural social capital" (Nahapiet & Ghoshal, 1998), influence knowledge creation. The configuration of an actor's set of direct ties has received some attention. Although the focus of this research is on triadic closure (i.e., whether or not an actor's partners are directly connected), two competing perspectives exist, each with different causal mechanisms linking network structure to innovation. One view argues disconnected networks increase creativity and innovation because they provide actors with timely access to diverse information² (Burt, 1992; 2004). An alternative view suggests dense networks, in which triads are closed and structural holes are absent, provide social capital because such structures generate trust, reciprocity norms and a shared identity, which lead to cooperation and knowledge sharing (Bourdieu, 1986; Coleman, 1988; Portes, 1998). Research has found support for both views, yielding conflicting results. While studies have found structural holes among a firm's partners enhance its knowledge creation (Baum et al., 2000; McEvily & Zaheer, 1999), other research shows network closure among alliance partners improves knowledge transfer and innovation (Ahuja, 2000; Dyer & Nobeoka, 2000; Schilling & Phelps, 2007). These studies highlight the benefits of network structure and largely overlook network composition. An examination of network composition may help resolve these conflicting findings and lead to a better understanding of how alliance networks influence firm innovation.

Although a fundamental benefit attributed to structural holes is timely access to diverse information (Burt, 1992), structural holes are neither a necessary nor a sufficient condition for

¹ Sampson (2007) examined the technological differences between firms in alliances and its affect on firm innovation. In contrast to what this study examines, Sampson (2007) did not investigate the technological diversity within a firm's network of alliance partners, the structure of this alliance network, or the exploratory content of a firm's innovations.

² While Burt (1992) also argued structural holes allow actors to behave free from the normative expectations of others in the network, research into the influence of network structure on innovation and creativity has stressed informational benefits, rather than control benefits, as the primary causal motor (e.g., Ahuja, 2000; Burt, 2004; Obstfeld, 2005).

such access (Reagans et al., 2004). The informational benefits contacts provide can be directly observed by examining the extent to which they specialize in different domains of knowledge (Reagans & McEvily, 2003; Rodan & Galunic, 2004). Observing differences in competencies also allows a more fine-grained measure of diversity than simply counting structural holes. Because competencies are stable and durable properties of firms (Patel & Pavitt, 1997), they are a compositional variable. Ties to partners with dissimilar knowledge stocks will provide a firm with access to diverse information and knowledge, independent of the structure of its local network. Thus, the benefits of network closure and access to diverse information and know-how can coexist. Research on interfirm alliances (Ahuja, 2000) and interpersonal networks (Reagans & McEvily, 2003; Rodan & Galunic, 2004) has shown that network density and knowledge diversity are empirically independent. However, alliance research has not examined the independent and interactive effects of network structure and network knowledge diversity on firm innovation.

A final limitation of research on alliance networks and firm innovation is that it largely ignores the novelty of the knowledge created and embodied in the innovations measured. Instead, studies have focused on the amount of innovation as indicated by survey items or counts of new products and patents. This approach implicitly assumes innovations are similar in their knowledge content. While research suggests firms typically pursue local search and produce exploitative innovations (e.g., Dosi, 1988; Martin & Mitchell, 1998), some research shows firms vary in the scope of their search and the exploratory content of their innovations (Ahuja & Lampert, 2001; Rosenkopf & Nerkar, 2001). A few studies have examined how organizational design decisions such as structure and vertical integration influence exploratory knowledge creation (Ahuja & Lahiri, 2006; Jansen et al., 2006; Sigelkow & Rivkin, 2005). However, research has not examined the effects of alliance network structure and network diversity on exploratory knowledge creation.

The purpose of this study is to address these limitations. I do so by examining the influence of the structure and composition of a firm's network of horizontal technology alliances on its degree of exploratory innovation. I focus on horizontal technology alliances (i.e., technical

linkages between firms in the same industry) for theoretical clarity. Exploratory innovation is the creation of technological knowledge that is novel relative to a firm's existing knowledge stock. While research often conceptualizes exploration as a process (March, 1991), the manifestation of this process can be observed by examining the exploratory content of a firm's innovations (Benner & Tushman, 2002; Jansen et al., 2006; Rosenkopf & Nerkar, 2001). An exploratory innovation embodies knowledge that differs from knowledge used by the firm in prior innovations and shows the firm has broadened its technical competence³ (Rosenkopf & Nerkar, 2001). Exploratory innovation is akin to firm-level radical innovation because such innovations embody knowledge outside a firm's extant technical competence (Dewar & Dutton, 1986; Greve, 2007).

Understanding the origins of exploratory innovation is significant for at least two reasons. First, strategy research suggests the creation of resources such as technological knowledge is the basis for sustainable competitive advantage (Teece et al., 1997). Second, exploratory innovations open new areas of technical advance, which provide the technological bases for new businesses and firm growth (Kim & Kogut, 1996). Thus, explaining the sources of exploratory innovation aids in understanding the sources of competitive advantage and firm growth.

I derive two predictions about the effect of horizontal technology alliances on firm exploratory innovation. First, in highlighting the role of network composition I draw on the recombinatory search literature (e.g., Fleming, 2001) and argue that access to a portfolio of technologically diverse partners increases the variance of search opportunities, resulting in greater exploratory innovation. Second, I note that while alliances provide access to diverse technical knowledge, they do not guarantee the effective transfer and integration of such knowledge.

Building on interorganizational social capital research (e.g., Ahuja, 2000; Inkpen & Tsang, 2005;

³ This conceptualization coincides with how exploitation and exploration have been defined and measured at the organization level of analysis (e.g., Benner & Tushman, 2002; Greve, 2007; Gupta et al., 2006; Jansen et al., 2006; March, 1991; McGrath, 2001; Schildt et al., 2005; Sidhu et al., 2007; Smith & Tushman, 2005; Stuart & Podolny, 1996) and the individual level (Audia & Goncalo, 2007). As He and Wong (2004: 485) concisely concluded, "exploration versus exploitation should be used with reference to a firm itself and its existing capabilities, resources and processes, not to a competitor or at the industry level." In other words, "one can only view acts of exploration or exploitation relative to a particular actor's vantage point" (Adner & Levinthal, 2008: 49).

Walker et al., 1997), I argue the extent to which a firm's partners are densely interconnected will generate structural social capital in the form of trust and reciprocity, which will increase a firm's ability to learn from diverse partners and enhance its exploratory innovation performance.

I test these predictions on a panel of 77 leading communications equipment manufacturers during 1987-1997 and find robust support for both hypotheses. This study contributes to the alliance and innovation literatures by addressing significant gaps in research on the influence of alliance networks on firm innovation. This is the first study of which I am aware that investigates the influence of alliance network structure and composition on firm exploratory innovation. The results show the technological diversity in a firm's alliance network and the density of the network increase exploratory innovation, independently and in combination. The results also suggest the presence of structural holes in a firm's network is not a necessary condition for providing it access to diverse information. The extent to which an actor's network is composed of alters with diverse knowledge bases will provide it access to diverse information, independent of network structure. The benefits of network closure and access to diverse information and know-how can coexist in a firm's alliance network and combining the two increases a firm's exploratory innovation.

THEORY AND HYPOTHESES

To understand when alliances influence a firm's production of exploratory innovations, I build on two complementary theoretical lenses: recombinatory search and social capital. The recombinatory search literature argues innovation is a problem-solving process in which solutions to economically valuable problems are discovered via search (Dosi, 1988). Search processes leading to the creation of new knowledge typically involve the novel recombination of existing elements of knowledge, problems, or solutions (Fleming, 2001; Nelson & Winter, 1982) or the reconfiguration of the ways in which knowledge elements are linked (Henderson & Clark, 1990). Search is uncertain, costly and guided by prior experience (Dosi, 1988).

Firms create knowledge by engaging in local and distant search (March, 1991). Local search, which is synonymous with exploitation, produces recombinations of familiar and well-

known knowledge elements, and is often the preferred mode of search (March, 1991; Stuart & Podolny 1996). In contrast, distant search, or exploration, involves recombinations of novel, unfamiliar knowledge and is often characterized by substantial costs and uncertainty (March, 1991; Nelson & Winter, 1982). Although distant search can be less efficient and less certain than local search, it increases the variance of search and the potential for highly novel or radical recombinations (Levinthal & March, 1981; Fleming, 2001; Rosenkopf & Nerkar, 2001).

While studies of innovation search have largely focused on where firms search for solutions, the interfirm learning literature has emphasized how firms search. This research argues that interfirm relationships are a mechanism for search and a medium of knowledge transfer (Ingram, 2002). Because knowledge is widely and heterogeneously distributed (von Hayek, 1945), the exchange of knowledge is a prerequisite for recombination (Nahapiet & Ghoshal, 1998). Firms that are able to search for and identify potentially useful elements of knowledge, conceive of how these knowledge components can be usefully recombined, and effectively access and assimilate this knowledge increase their knowledge creation (Galunic & Rodan, 1998). Alliances can play an important role in successful recombination by providing partners with access to each other's resources, thus increasing the amount and variety of knowledge available to a firm (Stuart, 2000).

Although alliances provide access to external sources of knowledge, they do not guarantee the effective detection, transfer and assimilation of this knowledge. These processes, and therefore the likelihood of successful recombination, are largely influenced by the social capital that inheres in a firm's alliance network (Inkpen & Tsang, 2005; Nahapiet & Ghoshal, 1998). Social capital refers to the instrumentally valuable, tangible and intangible resources that inhere in a network of social relationships among individuals, groups or organizations (Burt, 1992; Coleman, 1988; Portes, 1998). The extent to which social capital exists in a firm's network of horizontal technology alliances can increase its access to its partners' knowledge, the motivation of its partners to transfer knowledge, and the efficiency of knowledge exchange and transfer (Inkpen &

Tsang, 2005). I focus on structural social capital, which suggests the benefits of social ties are a function of the configuration of the ties in an actor's network (Nahapiet & Ghoshal, 1998).

All hypothesized relationships below are *ceteris paribus* arguments. That is, I expect the predicted effects to hold independent of the influence of other variables. Accordingly, in testing the hypotheses I control for observable and unobservable sources of heterogeneity that may be confounded with the hypothesized variables in order to isolate their unique, marginal effects.

Network technological diversity

Because exploratory innovations embody relatively novel (i.e., distant) knowledge, a necessary condition for firm exploratory innovation is access to dissimilar sources of knowledge (Greve, 2007; Jansen et al., 2006). Both the innovation search and individual creativity literatures emphasize the importance of access to diverse sources of knowledge for the creation of exploratory knowledge (Audia & Goncalo, 2007; Fleming, 2001; Greve, 2007; Nelson & Winter, 1982; Levinthal & March, 1981). An increase in the diversity of search increases the knowledge elements available for recombination, increasing combinatorial possibilities and the potential for novel recombinations (Fleming, 2001). Searching dissimilar knowledge sources also increases the variety of knowledge elements examined and the variance in the outcomes of search (Fleming, 2001). The "value of variance" (Mezias & Glynn, 1993) in distant search is that while it increases the number of failures it also increases the number of highly novel or exploratory solutions (Jansen et al., 2006; Levinthal & March, 1981). In contrast, individuals and organizations that exploit their established competences in their innovative problem-solving efforts typically experience more certain and immediate returns but produce incrementally innovative solutions that diverge little from their own prior solutions (Audia & Goncalo, 2007; Dosi, 1988; Fleming, 2001; Stuart & Podolny, 1996).

Searching diverse knowledge sources challenges existing cognitive structures and beliefs about cause-effect relationships and generates greater cognitive variety (Noteboom, 1999). This can reduce superstitious learning and stimulate deeper understanding of cause-effect relationships

(Levitt & March, 1988) and promote new associations, which often lead to highly novel insights and solutions (Simonton 1999). By searching diverse knowledge, firms can develop multiple, novel conceptualizations of problems and solutions and can potentially apply solutions from one domain to problems in another via analogies and metaphors that link these diverse domains (Gentner et al., 1997; Hargadon & Sutton 1997). Finally, searching diverse, nonredundant knowledge can stimulate intensive experimentation with new combinations, leading to breakthrough innovations (Ahuja & Lampert 2001; Phene et al., 2006; Quintana-Garcia & Benavides-Velasco, 2008).

Technological innovation involves tacit and socially embedded knowledge (Dosi, 1988). While technology is manifest in artifacts and institutionalized practices, it also includes ideas, beliefs, expectations, and practices of individuals involved in its development (Garud & Rappa, 1994). Technology is knowledge embedded in communities of practitioners (Layton, 1974), who, through experience, develop tacit understandings of how to solve problems related to its use and reproduction (von Hippel, 1988). Such knowledge is often embedded in organizational routines (Nelson & Winter, 1982). Firms attempting to integrate diverse technologies may lack the associated tacit knowledge and routines to do so. The specialized, tacit and embedded nature of technological knowledge makes market trading for it subject to severe exchange problems (Teece, 1992). Firms that can access such resources and overcome the difficulties in their transfer increase their likelihood of successful knowledge recombinations (Galunic & Rodan, 1998).

Strategic alliances are a primary means of accessing knowledge a firm does not have and can be an effective medium of knowledge transfer and integration (Hamel, 1991). Alliances provide a firm with direct and repeatable access to its partners' organizational routines, which reduces its ambiguity about a partner's knowledge and increases the effectiveness of its transfer and assimilation (Jensen & Szulanski 2007). Because of the increased social interaction and enhanced incentive alignment and monitoring features they provide, alliances are institutions better suited than market transactions for the repeated exchange of tacit, routine-embedded

knowledge (Teece, 1992). Thus, while alliances provide access to dissimilar technological knowledge, they can also facilitate its transfer and assimilation (Rosenkopf & Almeida, 2003).

Hypothesis 1: *The technological diversity in a firm's alliance network will be positively related to its subsequent degree of exploratory innovation.*

Structural social capital

Although an alliance provides access to a partner's knowledge, it does not guarantee the effective detection, transfer and assimilation of this knowledge (Hamel, 1991). The structural social capital derived from a dense alliance network complements a firm's ability to benefit from diverse partner knowledge by increasing the effective detection, transfer and assimilation of partner knowledge. Thus, network density will strengthen the influence of network diversity on firm exploratory innovation.

Structural social capital refers to the instrumentally valuable resources that inhere in a network of relationships; these resources are a function of the configuration of the ties in the network, independent of the properties of specific dyadic ties (Nahapiet & Ghoshal, 1998). In contrast, relational social capital is a property of a direct tie between actors (Nahapiet & Ghoshal, 1998). Two important forms of structural social capital are trust and reciprocity exchanges (Coleman, 1988; Portes, 1998). Both of these are products of dense networks (Portes, 1998).

Interfirm trust exists when a firm's personnel have confidence that a partner will not exploit their firm's vulnerabilities (Barney & Hansen, 1994). The high degree of connectivity in a dense network promotes trust among networked firms. Dense networks allow firms to learn about current and prospective partners through common third-party contacts, which reduce information asymmetries among firms and increases their "knowledge-based trust" in one another (Gulati et al., 2000). Network closure also promotes trust by increasing the costs of opportunism (Coleman, 1990). Opportunism is more costly in dense networks because it is readily discovered and rapidly communicated throughout the network, increasing the chances of costly sanctions (Coleman, 1990). Because a firm's behavior is more visible in a dense network, a single act of opportunism can damage its reputation, jeopardizing its existing alliances and diminishing future partnering

opportunities (Gulati, 1998; Kreps, 1990). Because acting opportunistically is potentially expensive, firms will refrain from doing so. Thus, dense networks also generate “enforceable” or “deterrence-based” trust ⁴ (Kreps, 1990; Raubb & Weesie, 1990; Portes, 1998).

Dense networks can produce another form of social capital – reciprocity exchanges – in which partners provide privileged access to resources because they expect recipients will repay them with something of roughly equivalent value (Coleman, 1988; Gouldner, 1960). A firm can encourage reciprocity between two of its partners by transferring reciprocal obligations owed to it by one partner to the other partner (Uzzi, 1997). Dense networks also encourage reciprocity exchanges by protecting relationships from opportunism (Portes, 1998). The sharing of privileged resources increases as actors become more confident that the associated obligations for repayment will eventually be met (Coleman, 1988; Macaulay, 1963).

The density of a firm’s alliance network will affect its exploratory innovation because the trust and reciprocity benefits of network closure reduce opportunism and promote knowledge sharing among partners. Opportunism can take many forms and is present to varying degrees in nearly all alliances (Luo, 2005). The risk of opportunism is pronounced in horizontal technology alliances due to partners’ competitive incentives and the inherent uncertainty in, and measurement and monitoring problems of, agreements involving the creation or exchange of technical knowledge (Pisano, 1989). Firms in such alliances are at risk of involuntary leakage of valuable knowledge, the withholding of effort and resources necessary for achieving alliance goals, the misrepresentation of newly discovered information and knowledge, and challenges in transferring tacit technical knowledge developed during the relationship (Gulati & Singh, 1998; Pisano, 1989).

⁴ A growing body of evidence provides direct and indirect empirical support for these arguments. Direct evidence that interfirm network closure reduces concerns of partner opportunism and/or promotes interfirm trust and cooperation is provided by qualitative research (Husted, 1994; Uzzi, 1996), simulation research (Buskens, 1998), and many large sample quantitative studies (Gulati & Sytch, 2008; Holm, Eriksson & Johanson, 1999; Robinson & Stuart, 2007; Noteboom, 1999; Rooks et al., 2000). Studies of interpersonal networks also provide direct evidence that network closure leads to more trusting and cooperative relationships (e.g., Berg, Dickhaut & McCabe, 1995; Buskens & Raub, 2002; Frank & Yasumoto, 1998; Gargiulo, 2003; Greif, 1993). Additional studies provide indirect evidence of the positive impact of interfirm network density on interfirm trust and cooperation (Ahuja, 2000; Bae & Gargiulo, 2004; Gulati & Gargiulo, 1999; Parkhe, 1993; Walker et al., 1997).

Trust and reciprocity act as informal safeguards of exchange, supplementing or mitigating the need for formal governance mechanisms (Granovetter, 1985; Powell, 1990). Informal governance reduces the threat of opportunism and increases firms' motivations to cooperate and share resources (Dyer & Singh, 1998). Trust reduces the extent to which alliance partners protect knowledge, increases their willingness to share knowledge, and increases interfirm learning and knowledge creation (Hamel, 1991; Kale et al., 2000; Larson, 1992). Reciprocity norms reinforce this willingness to share since firms can be confident their partners will reciprocate (Dyer & Nobeoka, 2000). As a result, the information and know-how shared will be less distorted, richer and of higher quality (Dyer & Nobeoka, 2000; Uzzi, 1997). Research suggests dense interfirm networks are better suited for transferring and integrating complex and tacit knowledge than networks with structural holes (Ahuja, 2000; Dyer & Nobeoka, 2000; Gilsing & Noteboom, 2006; Kogut, 2000). Thus, when innovation requires mobilizing and sustaining the transfer and development of complex tacit knowledge, dense networks are important (Obstfeld, 2005).

The trust and reciprocity benefits of network closure also facilitate exploratory innovation by promoting social interaction, experimentation, joint problem solving, triangulation and referrals. Trust and reciprocity promote intense interaction among personnel from partnered firms (Larson, 1992), improving the transfer of tacit and embedded knowledge (Zander & Kogut, 1995), which is critical to successful recombination (Galunic & Rodan, 1998). Intense interaction can also lead to the creation of partner-specific knowledge sharing routines that facilitate knowledge transfer (Lane & Lubatkin, 1998). The social capital associated with dense alliance networks encourages such relationship-specific investments (Walker et al., 1997). Structural social capital increases partners' joint problem solving and experimentation with different knowledge combinations, both of which can result in novel recombinations (Dyer & Nobeoka, 2000; Uzzi, 1997). Alliance partners provide alternative interpretations of technical problems and solutions, which enable a firm to compare, contrast and triangulate these perspectives (Nonaka, 1994). The rapid flow of information in dense networks provides firms with more opportunities to share and

expand their understanding of technical issues and can help establish a shared mode of discourse (Powell & Smith-Doerr, 2005). Finally, dense networks enable the quick identification of useful referrals, making it easier for firms to find potentially useful knowledge (Dyer & Nobeoka, 2000).

The structural social capital that network closure generates becomes more important for exploratory innovation as diversity increases. Increasing technological diversity among partners can increase exchange hazards and reduce effective cooperation. As the technological diversity in a firm's alliance network increases, the relative novelty of this knowledge, including the tacit portion of it, increases. This, in turn, increases partner uncertainty and ex ante and ex post contracting problems (Pisano, 1989). Technological dissimilarity also increases coordination problems (Sampson, 2004) and the potential for costly contractual renegotiations (Conner & Prahalad, 1996), both of which increase the risk of opportunism. Because technologically dissimilar firms have more to learn from each other, such firms may have stronger incentives to act opportunistically by expropriating partner knowledge (Hamel, 1991). Given these challenges to formal alliance governance, informal governance will become increasingly important in mitigating opportunism and promoting cooperation as technological diversity increases.

As the technological diversity within a firm's alliance network increases, it also becomes harder for the firm to learn from its partners (Cohen & Levinthal, 1990). The intense social interaction and joint problem solving that network closure promotes facilitates knowledge transfer, which is increasingly important as diversity grows. Trust and reciprocity can enhance a firm's willingness to 'teach,' which is more important for student firms as dissimilarity increases (Szulanski, 1996). At low levels of diversity, firms should find it easier to learn. The rapid diffusion of alternative interpretations of technical issues in dense networks is more valuable when partners are diverse because a variety of alternative perspectives increases the chances that at least some will be useful for a firm's recombination efforts (Nonaka, 1994). Technologically similar firms provide redundant interpretations of little value. The development of a shared perspective and technical discourse that density facilitates are more important when partners are diverse since

dissimilarity increases the variety of ways partners perceive and discuss technical issues (Kogut & Zander, 1996), making it more difficult for them to communicate with and learn from one another.

In sum, I expect network density will enhance the influence of network diversity on exploratory innovation. Indeed, in a qualitative study of six biotechnology firms, Maurer and Ebers (2006) found that dense external relations with diverse partners enabled these firms to access and integrate the resources needed for their continued growth and development.

Hypothesis 2: *The relationship between network diversity and firm exploratory innovation will be moderated by ego network density: as density increases, the effect of diversity on exploratory innovation will increase.*

RESEARCH METHODOLOGY

Sample and data

The research setting for this study is the global telecommunications equipment industry (SIC 366). Firms in this industry produce and market hardware and software that enables the transmission, switching and reception of voice, images, and data over both short and long distances using digital, analog, wireline and wireless technology. I chose this setting for two reasons. First, during the 1980s and 1990s this industry experienced significant changes in technology and competition, resulting in a growing use of technology alliances by incumbents (Amesse et al., 2004.). Second, since I use patent data I study an industry in which firms routinely and systematically patent their inventions (Hagedoorn & Cloudt 2003, Levin et al. 1987).

To minimize left and right censoring in data collection, I limited the sample period to 1987-1997. I limited the sample frame to public companies to ensure the availability and reliability of financial data. I limited the sample to the largest selling firms in the industry because complete and accurate alliance data are available for industry leaders more so than smaller firms (Gulati, 1995). To minimize survivor bias, I identified the top-selling firms in the industry at the beginning of the sample period. Following prescriptions for establishing network boundaries in empirical research (Laumann, Marsden, & Prensky, 1983), I restricted the network to both firms and alliances that focused on the telecommunications equipment industry. Recent alliance network

research has used similar network construction criteria (Rowley et al., 2000; Schilling & Phelps, 2007). These sampling criteria resulted in a sample of 77 firms headquartered in 13 countries.

I use patent data to assess the sample firms' technological knowledge because patents are valid and robust indicators of knowledge creation (Trajtenberg 1987). Knowledge creation is instantiated in inventions (Schmookler 1966). Patents are measures of novel inventions that are externally validated through the patent examination process (Griliches 1990). Because obtaining and maintaining IP protection is costly, a patent application represents a positive expectation by the inventor of the economic value of the invention (Griliches 1990). While patents measure only a codifiable portion of a firm's technical knowledge, they correlate with measures that incorporate tacit knowledge (Basberg 1982; Brouwer & Kleinknecht 1999). Patents are a reliable and valid measure of innovation in the telecom equipment industry (Hagedoorn & Cloudt, 2003).

I use U.S. patents, obtained from *Delphion*. Using patents from a single country maintains consistency, reliability and comparability across firms (Griliches 1990). U.S. patents are a good data source because of the rigor and procedural fairness used in granting them, the strong incentives firms have to obtain patent protection in the world's largest market, the high quality of services provided by the USPTO, and the U.S.'s reputation for providing effective IP protection (Pavitt 1988, Rivette 1993). I use the application date to assign granted patents to firms because this date more closely captures the timing of knowledge creation (Griliches 1990). Because patents are often assigned to subsidiaries, I carefully aggregated patents to the firm level⁵.

I used multiple sources for the collaboration data. I initially collected alliance data from the Securities Data Company (SDC) joint venture and alliance database. While the SDC database provided substantial alliance data, it suffers from many limitations⁶. I overcame these limitations

⁵ I identified all divisions, subsidiaries, and joint ventures of each sample firm (using *Who Owns Whom* and *The Directory of Corporate Affiliations*) as of 1980. I then traced each firm's history to account for name changes, division names, divestments, acquisitions, and joint ventures and obtained information on the timing of these events. This yielded a master list of entities that I used to identify all patents belonging to sample firms for the period of study.

⁶ First, SDC did not undertake systematic collection of alliance data until around 1989 (Anand & Khanna, 2000: 300). Second, SDC heavily relies on English language sources, resulting in a bias towards alliances involving partners from English-speaking countries. Third, SDC often includes announcements of alliances that were never formed. Fourth,

through systematic archival research using annual reports, 10K and 20F filings, *Moody's Industrial and International Manuals*, *Factiva*, *Lexis-Nexis* and *Dialog*. These last three databases index the historical full text of hundreds of business publications from all regions of the world and include articles translated to English from their original language and non-English publications⁷. I conducted broad key word searches to identify all instances of interfirm cooperation involving the sample firms. I recorded only collaborations that I could confirm using multiple sources. I examined around 1200 annual reports and SEC filings and over 180,000 electronic articles and printed over 8500 relevant news stories. Overall, the dataset from which this study draws includes 7904 alliances and 1967 acquisitions initiated during 1980-1996. I reviewed every record from the SDC data and corrected duplicate entries and other errors and omissions using secondary sources.

I collected firm attribute data using *Compustat*, annual reports, SEC filings, *The Japan Company Handbook*, *Worldscope* and *Global Vantage*.

--- Table 1 about here ---

Measurement: Dependent variable

Exploratory Innovation. Exploratory innovation is the creation of technological knowledge by a firm that is novel relative to its existing knowledge stock (Benner & Tushman, 2002; Rosenkopf & Nerkar, 2001). Following prior research (Audia & Goncalo, 2007; Benner & Tushman 2002, Katila & Ahuja, 2002; Rosenkopf & Nerkar 2001), I measure exploratory innovation using patent citation data. I began with the list of U.S. patent classes that correspond to the telecommunications equipment industry at the beginning of the sample period (see Table 1). I assessed the exploratory innovation of firm i in year t by classifying and tabulating all citations included in the firm's telecommunications equipment patents applied for in year t (and subsequently granted). I traced each citation to determine if the firm had used the same citation or if the citation was to a patent developed by the firm during the seven years before the focal year. I

SDC often records a single alliance multiple times, resulting in an artificial inflation of firm-alliance participation.

⁷ Non-English language articles and reports were examined by individuals fluent in the respective language using the list of key words. Instances of interfirm cooperation were identified and translated into English.

used a seven-year window because knowledge decays over time (Argote, 1999) and the intertemporal transfer of knowledge in firms is difficult (Nerkar, 2003). Thus, I classified each citation as new or used. I calculated exploratory innovation as: **Exploratory Innovations**_{it} = new citations_{it} / total citations_{it}. Because this measure is a share rather than a count of new citations, it captures a firm's propensity to produce exploratory innovations, independent of firm scale⁸.

The extent to which a firm reuses elements of knowledge (e.g., patent citations) it has prior experience with reflects that it is practicing local search and exploiting its extant knowledge stock. The extent to which it uses citations with which it has no prior experience is indicative of distant search and exploratory innovation (Benner & Tushman, 2002). Thus, this measure is a continuum ranging from pure exploitation (no exploration) at the low end to pure exploration (no exploitation) at the high end (Benner & Tushman, 2002). As such, this measure is consistent with research that conceptualizes and measures exploitation and exploration, or local and distant search, as ends of a continuum⁹ (Benner & Tushman, 2002; Greve, 2007; Schildt et al., 2005; Sidhu et al., 2007).

As a robustness check, I use an alternative measure of exploratory innovation used in prior research (Ahuja & Lahiri, 2006; Audia & Goncalo, 2007; Ahuja & Lampert, 2001; McGrath & Nerkar, 2004; Noteboom et al., 2007). I computed this measure as the number of new 3-digit technology classes in which firm *i* patented in year *t*. I classified a technology class as new if the

⁸ Two aspects of the patent data used to construct this measure merit discussion. First, the influence of a right-censoring bias, caused by the delay between patent application and issuance, is likely to be negligible in this study. Around 99% of all applications are reviewed within five years of application (Hall et al., 2001), which is the period between the end of the sample (1997) and the last year of patent data collection (2002). Second, patent examiners often add citations to patent applications (Alcacer & Gittelman, 2006), which suggests applicant firms are not necessarily aware of all cited patents. Research suggests examiners add citations with which assignee firms are familiar (Alcacer & Gittelman, 2006; Thompson, 2006). If such a bias was present, the primary measure of exploratory innovation would be biased downward, which would dampen the true effects of the hypothesized variables. Systematic relationships between unobserved examiner-added citations and the hypothesized variables (i.e., endogeneity) are highly unlikely. Endogeneity tests confirmed this (see Appendix). Third party citations will often manifest as noise in the measurement of patent-based variables (Jaffe, Trajtenberg, & Fogarty, 2002; Duguet & MacGarvie, 2005). Noise in the measurement of a dependent variable manifests in the regression residuals, which increases standard errors and reduces the likelihood of finding statistically significant effects (Gujarati, 1995).

⁹ While I focus on one domain of search (i.e., technological knowledge), firms search across multiple domains, such as customer and geographic space (Gupta et al., 2006; Sidhu et al., 2007). Portraying exploitation and exploration as ends of a continuum in one domain of search does not preclude the possibility that firms can simultaneously achieve high levels of both exploitation and exploration across multiple domains (Gupta et al., 2006). A universal argument about the mutual exclusivity or independence of exploitation and exploration may be impossible (Gupta et al., 2006).

firm had not patented in that class in the past seven years. The USPTO assigns patents to about 450 technology classes, with each class demarcating an area of technology. The extent to which a firm enters new technological domains is indicative of exploration (Ahuja & Lampert, 2001; McGrath & Nerkar, 2004). This measure is broader than the citation-based measure since it considers all technology classes in which a firm may patent.

Measurement: Independent variables

Following prior research (e.g., Ahuja, 2000; Stuart, 2000), I sample alliances involving technology development or exchange because my phenomenon of interest and theory are concerned with the transfer and creation of technological knowledge. I exclude unilateral licensing and alliances formed for the sole purpose of marketing, distribution or manufacturing.

Network Technological Diversity. To measure network technological diversity, I employ Rodan and Galunic's (2004) measure of knowledge heterogeneity. This measure incorporates information about the knowledge distance between a focal actor and each of its partners and the distance among the partners. I began at the dyad level and measured the technological distance between pairs of firms using Jaffe's (1986) index. For each firm-year, I measured the distribution of a firm's patents across primary patent classes. Following Sampson (2007), I used a moving four-year window to establish a firm's patenting profile. This distribution locates a firm in a multidimensional technology space, captured by a K -dimensional vector $f_i = (f_{i1} \dots f_{ik})$, where f_{ik} represents the fraction of firm i 's patents that are in patent class k . This approach assumes the distribution of a firm's patents across classes reflects the distribution of its technical knowledge (Jaffe, 1986). The technological distance, d , between firms i and j in year t was calculated as:

$$d_{ijt} = 1 - \left[\frac{\sum_{k=1}^K f_{ik}f_{jk}}{\left(\sum_{k=1}^K f_{ik}^2 \right)^{1/2} \left(\sum_{k=1}^K f_{jk}^2 \right)^{1/2}} \right]$$

This measure is bounded between 0 (complete similarity) and 1 (maximum diversity), and is symmetric for the two firms. I used these pairwise distance values to construct annual distance matrices, D_t , which reflect the technological distances between all possible pairs of sample firms.

Next, I computed the uniqueness of each partner j 's knowledge in firm i 's alliance network in year t . The uniqueness of a firm, j , is a function of the uniqueness of its partners, k , and firm j 's distance from them. Following Rodan and Galunic (2004), I define the uniqueness of firm j , u_j , as:

$$\lambda u_j = \sum_k d_{jk} * u_k .$$

The uniqueness of each firm is found in the solution of the eigen equation: $\lambda U = DU$, where U is an eigenvector of D and λ is its associated eigenvalue¹⁰. The elements of U are the uniqueness measures for each firm and D is the matrix of pairwise technological distances. I measure the technological diversity available to firm i in its (ego) network of alliance partners in year t as:

$$\text{Network Technological Diversity}_{it} = \frac{1}{N} \sum_{j=1}^N d_{ij} \lambda u_j ,$$

where d_{ij} is the technological distance of partner j from i and λu_j is j 's uniqueness score calculated for i 's N partners. The $1/N$ term compensates for the fact that lambda increases linearly with network size. This measure increases linearly with the distance among firm i and its partners and is a monotonically increasing function of network size (Rodan & Galunic, 2004).

To measure network structure, I constructed annual adjacency matrices for the period 1987-1996 that indicate the presence of a technology alliance, in existence at the end of the focal year, between all possible undirected pairwise combinations of sample firms. An alliance with more than two firms enters the adjacency matrix as separate dyadic combinations of all firms in the alliance. Of all sample alliances, 89% involve only two firms and the average alliance had 2.38 firms. Because alliances often endure longer than one year, constructing adjacency matrices using only alliances formed in the focal year would understate the true connectivity of the network. Data on both pre-sample alliance formation and alliance duration are needed to assess network structure accurately. I collected alliance data for each firm beginning in 1980 and researched each alliance to identify its date of dissolution or continuance through the last sample year¹¹.

¹⁰ Lambda (λ) is a constant required so the equations have a nonzero solution. To assure these uniqueness values are nonnegative, the largest eigenvalue (λ) was used (see Bonacich, 1987).

¹¹ I researched each alliance using the sources described previously. I also contacted company personnel to identify

Density: Ego network density is measured by disregarding ego's direct relationships and is computed as the percentage of all possible ties that exist among ego's alters (Scott, 1991). Ego networks in which a firm's alliance partners are themselves allied imply higher values of *Density*, while a network rich in structural holes would yield a lower value. As robustness tests of the effect of *Density*, I substitute Burt's (1992) measures of *Efficiency* and then *Constraint* in alternative specifications (see the Appendix). Both *Efficiency* and *Constraint* are measures of triadic closure (see Borgatti, 1997 for a comparison of these three measures). Figure 1 presents an example of an ego network for a sample firm. The figure shows Motorola's network of technology alliances as of the end of 1992 and provides the values for the Density, Efficiency and Constraint of this network.

--- Figure 1 about here ---

Control variables

To minimize alternative explanations and to assess the marginal effects of the hypothesized variables, I control for several firm- and alliance-level variables that may influence exploratory innovation and that may be confounded with the explanatory variables. Because this study is focused on the firm level of analysis, I aggregate alliance-level observations to the firm level.

Network Size. More alliance partners may provide a firm with access to greater technical diversity. Moreover, measures of ego network density are sensitive to network size, making network size an important control variable (Friedkin, 1981). I compute *Network Size* as the number of telecommunications technology alliance partners maintained by firm *i* in year *t*.

Duration. Older alliances can lead to greater interfirm trust (Gulati, 1995), stronger reciprocity norms (Larson, 1992), a shared identity (Larson, 1992), and relationship-specific

dissolution dates, which proved very useful in identifying the termination or on-going status of joint ventures. For nearly all JVs, I was able to identify the month of termination or if it was on-going at the end of the sample period. For the remaining JVs, I assumed each existed until the end of the last year in which it was documented or until the end of the year after the year it was founded, whichever was later. For non-JV alliances, I recorded termination based on specified tenure, if mentioned in the archival sources, or on announcement of dissolution (either from archival sources or company contact). In cases where I could not establish precise dissolution, I followed Ahuja (2000) and presumed an alliance to exist until the end of the last year in which it was documented or until the end of the year after the year it was founded, whichever was later. I performed a t-test of the difference in mean duration between alliances with formal dissolution announcements and those using assumed dissolution dates and found no significant difference.

routines (Levinthal & Fichman, 1988), leading to greater interfirm learning (Simonin, 1999). I measure *Duration* as the average number of years firm *i* has been involved in its existing telecommunications technology alliances as of the end of year *t* (see footnote 11 above).

Repeated Ties. Prior ties between firms can increase interfirm trust (Gulati, 1995), the development of relation-specific learning heuristics and interfirm learning (Lane & Lubatkin, 1998). Following Gulati and Gargiulo (1999), I calculate *Repeated Ties* as the average number of alliances firm *i* formed with its current group of alliance partners in the five years prior to year *t*.

JV. Research suggests equity joint ventures are superior mechanisms for interfirm learning and knowledge transfer (Dutta & Weiss, 1997; Kogut, 1988). Empirical evidence supports this prediction (Mowery et al., 1996). I compute *JV* as the proportion of firm *i*'s portfolio of telecommunications technology alliances governed by equity joint venture in year *t*.

International. International alliances tend to experience greater coordination and communication problems and cultural conflicts, which can reduce cooperation, information flows and interfirm learning (Kotabe & Swan, 1995; Lyles & Salk, 1996). International alliances also provide firms access to diverse knowledge (Rosenkopf & Almeida, 2003). I measure *International* as the proportion of firm *i*'s telecom technology alliances in year *t* that involved foreign partners.

Market Overlap. Because partners tend to protect their knowledge when they are product-market competitors, overlaps in partners' markets can impede interfirm knowledge transfer (Dutta & Weiss, 1997). I compute *Market Overlap* as the proportion of firm *i*'s portfolio of telecom technology alliances in year *t* having partners with the same primary four-digit SIC as firm *i*.

Sales. Prior research has proved inconclusive in determining whether small or large firms are more innovative (Cohen & Levin, 1989). This lack of conclusiveness may be due to the presence of both negative and positive effects of size on firm innovation (Teece, 1992). I control for the effect of firm size using the natural log of sales (reported in \$US million) for firm *i* year *t*.

Current. The availability of slack resources can increase exploratory search (Singh, 1986) and lead to greater innovative performance (Nohria & Gulati, 1996). I control for the unabsorbed

slack resources of firm i in year t using its current ratio (Singh, 1986).

R&D. A firm's R&D expenditures are investments in knowledge creation (Griliches, 1990) and contribute to its ability to absorb extramural knowledge (Cohen & Levinthal, 1990). I control for R&D using the natural log of R&D expenditures (recorded in \$US million) of firm i in year t .

Patent Stock. The greater the number of patents a firm has, the greater the number of patents and references the firm has to cite, which may result in a negative effect on the citation-based measure of exploratory innovation. A firm's patent stock is also a good indicator of the depth of its technological resources and absorptive capacity (Henderson & Cockburn, 1994). I control for the number of firm i 's patents obtained in the four years prior to and including year t .

Firm Age. As firms age, they tend to exploit their existing technological competencies rather than explore new and unfamiliar technologies (Sorensen & Stuart, 2000). I operationalize firm age as the number of years from the date of founding of firm i to year t .

Alliance Experience. Prior alliance experience enhances the collaborative capability of a firm, which can result in improved learning and knowledge transfer from alliance partners (Sampson, 2005). I measure alliance experience as the total number of alliances, regardless of value chain or industrial activity, formed by firm i in the seven years before year t .

Firm Technological Diversity. Technologically diverse firms may be more innovative due to greater internal knowledge flows (Garcia-Vega, 2006) and more able to absorb extramural knowledge (Cohen & Levinthal 1990). I measure firm i 's technological diversity in year t using Hall's (2002) adjusted Herfindahl index:

$$\text{Firm Technological Diversity}_{it} = [1 - \sum_{j=1}^J \left(\frac{N_{jit}}{N_{it}}\right)^2] * \frac{N_{it}}{N_{it} - 1} ,$$

where N_{it} is the total number of patents obtained by firm i in the four years prior to and including year t . N_{jit} is the number of patents in primary technology class j in firm i 's four year patent stock. This variable may take on values between 0 (no diversity) to 1 (maximum diversity).

Acquisitions. Acquisitions can enhance acquirer innovation (Ahuja & Katila, 2001).

Because firms often use both acquisitions and alliances to source knowledge (Arora & Gambardella, 1990), the effect of alliances on exploratory innovation may be confounded with that of acquisitions. I control for the number of telecom equipment acquisitions (i.e., where the target company's primary SIC = 366) by firm i during the four years prior to and including year t .

USCan/Europe/Asia. Firms from different regions vary in their innovation performance (Cohen & Levin, 1989). I use regional dummies to indicate a firm's origin. *USCan* is coded 1 when the firm is headquartered in the U.S. or Canada and *Europe* is coded 1 when the firm is headquartered in Europe. *Asia* includes Japan and South Korea and is the omitted category.

Model Specification and Estimation

The dependent variable is a proportion, which presents several challenges to linear regression (Gujarati, 1995). Thus, I use three alternative modeling approaches. First, I estimate the models with Exploratory Innovation as the dependent variable using panel linear regression and robust standard errors. Following standard econometric practice (Greene, 1997), I also estimate models with a log-odds transformation of Exploratory Innovation¹². Finally, I follow Papke and Wooldridge (2005) and estimate models using a generalized estimating equation (GEE) approach (Liang & Zenger, 1986), in which I specify a probit link function and an exchangeable correlation matrix and compute robust errors. As a robustness check, I compare the results from these alternative specifications. I include year dummies to control for systematic period effects such as differences in macroeconomic conditions or industry technological opportunity. To control for unobserved firm effects, such as differences in firms' motivations to pursue and abilities to develop exploratory innovations, I use both fixed and random effects. Because the use of random effects assumes errors and regressors are uncorrelated, I use a Hausman (1978) test to test this assumption. I also check for first-order serial autocorrelation. All independent variables are lagged one year, which reduces concerns of reverse causality and avoids simultaneity.

¹² The transformed variable is: $\ln(\text{Exploratory Innovation}/1 - \text{Exploratory Innovation})$. Because the transformation is undefined when Exploratory Innovation is equal to 0 or 1, I recoded these values as follows: 0=0.0001 and 1=0.9999.

--- Tables 2 & 3 about here ---

RESULTS

Table 2 reports descriptive statistics and correlations. The panel is unbalanced and consists of 77 firms and 707 firm-year observations. Table 3 presents the results of the panel regression analysis used to test the hypotheses. In computing the interaction term I first mean-centered the lower-order component variables to minimize multicollinearity (Jaccard & Turrisi, 2003). I report the results for untransformed Exploratory Innovation for ease of interpretation. The results using a logit transformation and from GEE estimation are consistent with those reported in Table 3. I estimated Models 1-4 using firm random effects for three reasons: (1) significant unobserved heterogeneity was present, (2) Hausman (1978) specification tests were not significant, supporting the use of random effects, and (3) significant serial correlation was not present. The Wald statistics at the bottom of Table 3 indicate models 2-4 provide significant improvement in fit relative to model 1. Table 3 presents robust standard errors and all significance levels are for two-tailed tests.

Hypothesis 1 predicted a positive effect of network technological diversity on firm exploratory innovation. Models 2-4 in Table 3 provide support for this hypothesis. In each of these models, Network Technological Diversity_{it-1} exhibited a positive and significant effect on exploratory innovation. Hypothesis 2 predicted that ego network density strengthens the effect of network technological diversity on exploratory innovation. Model 4 shows the interaction had a significant positive effect on exploratory innovation, supporting hypothesis 2. Unexpectedly, density had a positive and significant effect on exploratory innovation, independent of diversity (Models 3-4). In sum, these results provide consistent support for both H1 and H2. An assessment of the robustness of the results and alternative explanations is provided in the Appendix.

DISCUSSION

This study was motivated by important limitations of research on alliance networks and firm innovation. This research has largely ignored the potential influence of network composition; particularly the technological diversity of a firm's partners. This research also draws on seemingly

incompatible theoretical arguments and has produced conflicting empirical results regarding the influence of network structure. These conflicts stem from an assumption that a firm's access to diverse information and the innovation benefits of network closure are mutually exclusive. Due in part to this assumption, potential complementarities between network structure and composition have been largely unexamined. Finally, research on alliance networks and firm innovation has focused on the volume of firm innovation, with little consideration of its exploratory content.

This study addressed these limitations by examining the influence of the composition and structure of a firm's local network of horizontal technology alliances on its degree of exploratory innovation. The theoretical framework suggested network composition and structure play different, yet complementary roles in exploratory innovation. Regarding network composition, I drew on research on recombinatory search to suggest access to diverse knowledge via alliances increases the variance of search, thus increasing firm exploratory innovation. Regarding network structure, I built on the social capital literature and argued the density of a firm's horizontal alliance network increases its ability to access, mobilize, and integrate its partners' knowledge, thus increasing its ability to benefit from technologically diverse partners. In so doing, this study moved beyond the dyadic perspective typically used in interorganizational learning research (cf. Tiwana, 2008) and examined characteristics of a firm's portfolio of interfirm relationships.

In sum, the results are consistent with the predictions of the theoretical framework. I found network-level technological diversity had a positive effect on exploratory innovation¹³. I also found the density of a firm's network of horizontal technology alliances strengthened the effect of diversity. Unexpectedly, I found network density had a positive main effect on exploratory innovation. I speculate on this result below. These results do not seem to be biased by endogeneity and are robust to the use of several firm and alliance level controls, alternative specifications and estimation routines, firm fixed and random effects, and the use of alternative measures (see

¹³ In alternative model specifications, I tested for the presence of a non-linear effect of network diversity by including the square of this measure. In all estimations, this squared term was not statistically significant (see Appendix).

Appendix). This study has important implications for research and practice.

This study contributes to a debate in the literature concerning the network structure of social capital by suggesting that research has overemphasized the informational benefits of structural holes for firm innovation. Prior research assumes structural holes increase an actor's timely access to diverse information. Because structural holes and network closure are inversely related, this argument implies the informational benefits of structural holes must come at the expense of the benefits of network closure, and vice versa. Prior conflicting findings about the effect of structural holes on firm innovation may be influenced by a confounding of the structural holes effect with an unobserved compositional effect of partner knowledge diversity. This study suggests the extent to which an actor's network is composed of alters with diverse knowledge bases will provide it access to informational diversity, independent of network structure. The benefits of network closure and access to diverse information and know-how can co-exist in a firm's alliance network and the combination of the two enhances its exploratory innovation.

This study also contributes to the innovation search literature and an understanding of innovation. Much of this literature stresses the proclivity of firms to practice local search. Little research explores how firms are able to overcome the inertial tendencies of local search. The results of this study suggest having access to diverse knowledge is important. This finding reinforces the results of recent alliance-level research, which shows partner technological diversity affects the rate of firm innovation (Sampson, 2007). It is also consistent with research that shows that alliances enable knowledge transfer between technologically different partners (Rosenkopf & Almeida, 2003). While research has shown alliances enhance firm innovation performance, it is difficult to establish from these results whether firms expanded their technical competencies in the process. The findings of this study suggest alliances can spur exploratory innovation when they provide access to technologically diverse partners that are densely connected to one another.

An unexpected finding of this study was a positive main effect of alliance network density on firm exploratory innovation. This result is surprising because it suggests alliance network

closure directly enhances a firm's exploratory innovation, independent of the influence of network diversity. This result might be explained by considering the relationship between social control and network structure. Because informational diversity is a correlate of structural holes, once the influence of such diversity is controlled, the direct effect of the social control aspect of network structure can be isolated and observed. Structural holes theory argues the absence of social control, which structural holes provide, is instrumentally beneficial for actors (Burt, 1992). In contrast, the logic of network closure suggests the presence of social control is beneficial to network members because it reduces opportunism and increases cooperation (Coleman, 1990). The results of this study suggest the social control (i.e., informal governance) associated with dense horizontal alliance networks directly benefits a firm's exploratory innovation. Network closure provides informal governance that promotes knowledge sharing, intense social interaction, experimentation and joint problem solving among alliance partners. This may lead highly similar partners to interactively experiment with combining knowledge in ways they had not previously explored and discover highly novel combinations. Gilsing and Noteboom (2006) provide qualitative evidence consistent with this explanation. Given the substantial risk of opportunism in horizontal technology alliances, such recombinations may only be realized when the social control generated by network closure encourages partners to intensely cooperate. Thus, where the threat of opportunism is pronounced and cooperation is difficult to achieve, such as in horizontal technology alliances, network closure can directly enhance firm innovation (Ahuja, 2000).

An implication of this speculative explanation is that the value of network structure is contingent on both task and context (Adler & Kwon, 2002; Ahuja, 2000). The measures of exploratory innovation and density used in this study are unidimensional and range, respectively, from pure exploitation to pure exploration and from maximum efficiency to maximum density. Thus, the results suggest, after controlling for the effect of network diversity, dense networks of horizontal technology alliances enhanced exploratory innovation while sparse networks impeded it. Because the social control effect of network closure is independent of the effect of knowledge

diversity, I speculate that when the task of developing exploratory innovations in the context of horizontal technology alliances is important, dense networks are likely to be more beneficial than sparse networks. In contrast, when the efficient exploitation of existing technological knowledge is the goal and the context is collaboration among non-competitors, networks with structural holes may be advantageous. This represents an intriguing opportunity for future research.

The results of this study also have managerial implications. The findings confirm alliances can improve a firm's development of exploratory innovations. The theory and results point to the benefits of forming alliances with technologically diverse partners in densely connected networks. This study suggests managers should attend to the structure of the alliance networks in which their firms are embedded because these structures have implications for firm performance. While technology alliance partners are often selected based on an evaluation of their technological capabilities (Stuart, 1998), the results of this study suggest that a firm's ability to effectively learn from technologically diverse partners depends on the degree of network closure around these relationships¹⁴. Thus, managers should evaluate how their choices of forming new alliances and ending existing relationships will affect the structure of their networks. Moving from the dyad level of analysis to the network level can sensitize managers to the importance of understanding how social structure influences firm performance (Parise & Casher, 2003).

Although promising, the results of this study should be considered in light of its limitations. First, because I used patents to assess exploratory innovation, the measure may not capture all of a firm's exploratory innovations. To the extent firms are systematically biased in patenting explorative knowledge for unobserved reasons, parameter estimates may be biased. I attempted to control for this potential source of bias using control variables and firm effects. Second, firms may patent knowledge in anticipation of entering alliances because of concerns about future leakage of this knowledge to partners (Brouwer & Kleinknecht, 1999). Exploratory

¹⁴ I thank an anonymous reviewer for suggesting this implication.

inventions tend to have a greater impact on subsequent technological development (Rosenkopf & Nerkar, 2001) and may therefore be of greater economic value (Narin et al., 1987). Thus, firms may patent exploratory inventions before entering alliances to appropriate their greater economic value. The findings of this study may be influenced by such unobserved appropriation concerns. The use of a one-year lag between collaboration and patenting reduces the likelihood of such a bias. Firm effects also mitigate this bias by controlling for such unobserved heterogeneity.

Another possible limitation is that an alliance survivor bias may have influenced the results. If sample firms formed alliances with the intent of exploratory learning and if successful alliances survived (i.e., firms systematically selected out of unbeneficial alliances), then observed alliances will be those that yielded the greatest exploratory benefit. Such a self-selection process may manifest as endogeneity due to omitted variables, resulting in an upward bias of alliance-related effects. Such a bias is probably negligible in this study. First, I control for omitted variables using both random and fixed effects. Second, because I have time-varying data on alliances and I observe both alliance formation and dissolution, my data include both successful and unsuccessful alliances. Third, research shows firms often exit alliances before they yield knowledge transfer benefits (Deeds & Rothaermel, 2003) and often maintain alliances that negatively affect interfirm knowledge transfer (Gomes-Casseres et al., 2006). Fourth, firms enter technology alliances for reasons other than technological exploration (Hagedoorn, 1993). Finally, if alliances that are beneficial for exploration tend to survive, I would expect a positive effect of the number of alliances maintained by a firm on its exploratory innovation. I do not observe such an effect in this study. Thus, I believe an alliance survivor bias does not unduly affect my results.

A final limitation concerns the generalizability of the findings. The results may be unique to the time period studied, the sampled firms, or the industry context. Further corroboratory evidence using data from different periods, samples and industry contexts is needed to externally validate this study's findings.

The discussion of this study's results suggests an additional opportunity for future

research. While the results are consistent with theoretical expectations, a better understanding of the micro-sociological foundations that underlie the observed effects of alliance network structure and composition is needed to validate the causal inferences of this study. Much of the inter-organizational network literature draws on research on interpersonal networks. Future research needs to examine the extent to which causal explanations from interpersonal research are isomorphic to the interorganizational level. Longitudinal qualitative research should explore how interorganizational and interpersonal networks interact to produce social capital and how this social capital influences knowledge transfer and innovation.

REFERENCES

- Adler, P.S. & Kwon, S.K. 2002. Social capital: Prospects for a new concept. *Academy of Management Review*, 27: 17-40.
- Ahuja, G. 2000. Collaboration networks, structural holes and innovation: A longitudinal study. *Administrative Science Quarterly*, 45: 425-455.
- Ahuja, G., & Katila, R. 2001. Technological acquisitions and the innovation performance of acquiring firms: A longitudinal study. *Strategic Management Journal*, 22: 197-220.
- Ahuja, G. & Lahiri, N. 2006. Does vertical integration impact technological search? Evidence from the semiconductor industry. Working Paper.
- Ahuja, G. & Lampert, C.M. 2001. Entrepreneurship in the large corporation: A longitudinal study of how established firms create breakthrough inventions. *Strategic Management Journal*, 22: 521-543.
- Alcacer, J., & Gittelman, M.. 2006. How do I know what you know? Patent examiners and the generation of patent citations. *Review of Economics and Statistics*, 88 (4): 774-779.
- Amesse, F., Latour, R., Rebolledo, C., & Séguin-Dulude, L. 2004. The telecommunications equipment industry in the 1990s: from alliances to mergers and acquisitions. *Technovation*, 24: 885-897.
- Anand, B.N. & Khanna, T. 2000. Do firms learn to create value? The case of alliances. *Strategic Management Journal*, 21: 295-315.
- Arora, A., & Gambardella, A. 1990. Complementarities And External Linkages: The strategies of large corporations in biotechnology. *The Journal of Industrial Economics*, 38(4): 361-379.
- Argote, L. 1999. *Organizational Learning: Creating, Retaining, and Transferring Knowledge*. Boston: Kluwer Academic Publishers.
- Badaracco, J.L. 1991. *The Knowledge Link: How Firms Compete through Strategic Alliances*. Boston: Harvard Business School Press.
- Bae, J. & M. Gargiulo, M. 2004. Partner substitutability, alliance network structure, and firm profitability in the telecommunications industry. *Academy of Management Journal*, 47: 843–859.
- Barney, J.B. & Hansen, M.H. 1994. Trustworthiness as a source of competitive advantage. *Strategic Management Journal*, 15: 175-190.
- Baum, J.A.C., Calabrese, T., & Silverman, B.S. 2000. Don't go it alone: Alliance network composition and startups' performance in Canadian biotechnology. *Strategic Management Journal*, 21: 267-294.

- Benner, M.J. & Tushman, M. 2002. Process management and technological innovation: A longitudinal study of the paint and photography industries. *Administrative Science Quarterly*, 47: 676-698.
- Berg, J., Dickhaut, J. & McCabe, K. 1995. Trust, reciprocity, and social history. *Games and Economic Behavior*, 10: 122-142.
- Bonacich, P. 1987. Power and centrality: A family of measures. *American Journal of Sociology*, 92: 1170-1182.
- Borgatti, S.P. 1997. Structural holes: Unpacking Burt's redundancy measures. *Connections*, 20(1): 35-38.
- Bourdieu, P. 1986. The forms of capital. In J.G. Richardson (Ed.), *Handbook of theory and research for the sociology of education* (pp. 241-258). Westport, CT: Greenwood Press.
- Brouwer, E. & Kleinknecht, A. 1999. Innovative output and a firm's propensity to patent. An exploration of CIS micro data. *Research Policy*, 28: 615-624.
- Burt, R.S. 1992. *Structural Holes*. Cambridge, MA: Harvard University Press.
- Burt, R.S. 2004. Structural holes and good ideas. *American Journal of Sociology*, 110: 349-399.
- Buskens, V. & Raub, W. 2002. Embedded trust: Control and learning. *Group Cohesion, Trust and Solidarity*, 19: 167-202.
- Cohen, W.M. & Levinthal, D.A. 1990. Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35, 128-152.
- Coleman, J.S. 1988. Social capital in the creation of human capital. *American Journal of Sociology*, 94 (supplement): S95-S120.
- Coleman, J.S. 1990. *Foundations of social theory*. Cambridge, MA: Belknap Press of Harvard University Press.
- Conner, K.K. & Prahalad, C.K. 1996. A resource based theory of the firm: knowledge versus opportunism. *Organization Science*, 7: 477: 501.
- Cyert, R. & March, J. 1963. *A behavioral theory of the firm*. Englewood Cliffs, NJ: Prentice-Hall.
- Davidson, R. & MacKinnon, J.G. 1993. *Estimation and inference in econometrics*. New York: Oxford University Press.
- Deeds, D.L. & Rothaermel, F.T. 2003. Honeymoons and liabilities: The relationship between age and performance in research and development alliances. *Journal of Product Innovation Management*, 20: 468-484.

Dewar, R.D. & Dutton, J.E. 1986. The adoption of radical and incremental innovations: An empirical analysis. *Management Science*, 32: 1422-1433.

Dosi, G. 1988. Sources, procedures, and microeconomic effects of innovation. *Journal of Economic Literature*, 26: 1120-1171.

Duguet, E. & MacGarvie, M. 2005. How well do patent citations measure knowledge spillovers? Evidence from French innovative surveys. *Economics of Innovation and New Technology*, 14(5): 375-393.

Dutta, S. & Weiss, A.M. 1997. The relationship between a firm's level of technological innovativeness and its pattern of partnership agreements. *Management Science*, 43, 343-356.

Dyer, J.H. & Singh, H. 1998. The relational view: Cooperative strategy and sources of interorganizational competitive advantage. *Academy of Management Review*, 23, 660-679.

Dyer, J.H. & Nobeoka, K. 2000. Creating and managing a high-performance knowledge-sharing network: The Toyota case. *Strategic Management Journal*, 21: 345-367.

Fleming, L. 2001. Recombinant uncertainty in technological search. *Management Science*, 47: 117-132.

Frank, K.A. & Yasumoto, J.Y. 1998. Linking action to social structure within a system: Social capital within and between subgroups. *American Journal of Sociology*, 104: 642-686.

Friedkin, N.E. 1981. The development of structure in random networks: An analysis of the effects of increasing network density on five measures of structure. *Social Networks*, 3: 41-52.

Galunic, D.C. & Rodan, S. 1998. Resource recombinations in the firm: Knowledge structures and the potential for Schumpeterian innovation. *Strategic Management Journal*, 19: 1193-1201.

Garcia-Vega, M. 2006. Does technological diversification promote innovation? *Research Policy*, 35(2): 230-246.

Gargiulo, M. 2003. Network closure and third party cooperation. INSEAD working paper.

Garud, R., & Rappa, M. 1994. A socio-cognitive model of technology evolution. *Organization Science*, 5: 344-362.

Gentner, D., Brem, S., Ferguson, R., Markman, A., Levidow, B. B., Wolff, P., & Forbus, K. 1997. Analogical reasoning and conceptual change: A case study of Johannes Kepler. *The Journal of the Learning Sciences*, 6: 3-40.

Gilsing, V. & Noteboom, B. 2006. Exploration and exploitation in innovation systems: The case of pharmaceutical biotechnology. *Research Policy*, 35: 1-23.

- Gomes-Casseres, B., J. Hagedoorn & A. Jaffe. 2006. Do alliances promote knowledge flows? *Journal of Financial Economics*, 80: 5-33.
- Gouldner, A.W. 1960. The norm of reciprocity: A preliminary statement. *American Sociological Review*, 25: 161-178.
- Granovetter, M.S. 1985. Economic action and social structure: The problem of embeddedness. *American Journal of Sociology*, 91: 481-510.
- Greene, W.H. 1997. *Econometric Analysis*, 3rd edition. Upper Saddle River, NJ: Prentice Hall.
- Greif, A. 1993. Contract enforceability and the economic institutions in early trade: the Maghribi traders' coalition, *American Economic Review*, 83: 525-548.
- Greve, H.R. 2007. Exploration and exploitation in product innovation. *Industrial and Corporate Change*, 16: 945-975.
- Griliches, Z. 1990. Patent statistics as economic indicators: A survey. *Journal of Economics Literature*, 28: 1661-1707.
- Gujarati, D.N. 1995. *Basic Econometrics*, 3rd edition. New York: McGraw-Hill.
- Gulati, R. 1995. Does familiarity breed trust? The implications of repeated ties for contractual choice in alliances. *Academy of Management Journal*, 38: 85-112.
- Gulati, R. 1998. Alliances and networks. *Strategic Management Journal*, 19: 293-317.
- Gulati, R. 2007. *Managing network resources: Alliances, affiliations and other relational assets*. Oxford: Oxford University Press.
- Gulati, R. & Singh, H. 1998. The architecture of cooperation: Managing coordination costs and appropriation concerns in strategic alliances. *Administrative Science Quarterly*, 43: 781-814.
- Gulati, R. & Gargiulo, M. 1999. Where do interorganizational networks come from? *American Journal of Sociology*, 104: 1439-1493.
- Gulati, R., Nohria, N. & Zahra, A. 2000. Strategic networks. *Strategic Management Journal*, 21: 203-215.
- Gulati, R. & Sytch, M. 2008. Does familiarity breed trust? Revisiting the antecedents of trust. *Managerial and Decision Economics*, 29: 165-190.
- Gupta, A.K., Smith, K.G., & Shalley, C.E. 2006. The interplay between exploration and exploitation. *Academy of Management Journal*, 49: 693-706.

- Hagedoorn, J. 1993. Understanding the rationale of strategic technology partnering: Interorganizational modes of cooperation and sectoral differences. *Strategic Management Journal*, 14: 371-385.
- Hagedoorn, J. & Cloudt, M. 2003. Measuring innovative performance: is there an advantage in using multiple indicators? *Research Policy*, 32: 1365-1379.
- Hall, B. H. 2002. A note on the bias in the Herfindahl based on count data. In A. Jaffe and M. Trajtenberg (Eds.), *Patents, Citations, and Innovation*: 149-156 Cambridge, MA: MIT Press.
- Hall, B.H., Jaffe, A.B., & Trajtenberg, M. 2001. The NBER patent citations data file: Lessons, insights, and methodological tools. NBER Working Paper 8498.
- Hargadon, A., & Sutton, R.I. 1997. Technology brokering and innovation in a product development firm. *Administrative Science Quarterly*, 42: 716-749.
- Hamel, G. 1991. Competition for competence and inter-partner learning within international strategic alliances. *Strategic Management Journal*, 12(summer special issue): 83-103.
- Hausman, J. 1978. Specification tests in econometrics. *Econometrica*, 46:1251-1271.
- Hausman, J., Hall, B., & Griliches, Z. 1984. Econometric models for count data with an application to patents-R&D relationship. *Econometrica*, 52: 909-938.
- He, Z.L. & Wong, P.K.. 2004. Exploration versus exploitation: An empirical test of the ambidexterity hypothesis. *Organization Science*, 15: 481-494.
- Henderson, R. M. & Clark, K.B. 1990. Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, 35: 9-30.
- Holm, D.B., Eriksson, K. & Johanson, J. 1999. Creating value through mutual commitment to business network relationships. *Strategic Management Journal*, 20: 467-486.
- Huber, G.P. 1991. Organizational learning: The contributing processes and literatures. *Organization Science*, 2: 88-115.
- Husted, B.W. 1994. Transaction costs, norms, and social networks. *Business and Society*, 33: 30-44.
- Ingram, P. 2002. Interorganizational Learning. In J.A.C. Baum (ed.), *Companion to Organizations*. Blackwell, New York, 642-663.
- Inkpen, A.C. & Tsang, E.W.K. 2005. Social capital, networks, and knowledge transfer. *Academy of Management Review*, 30: 146-165.

Jaccard, J. & Turrisi, R. 2003. ***Interaction Effects in Multiple Regression***. 2nd edition. Thousand Oaks, CA: Sage.

Jaffe, A. 1986. Technological opportunity and spillovers of R&D: Evidence from firms' patents, profits and market value. ***American Economic Review***, 76: 984-1001.

Jaffe, A. B., Trajtenberg, M., & Fogarty, M. S. 2002. The meaning of patent citations: report on the NBER/Case-Western Reserve survey of patentees. In A. B. Jaffe, and M. Trajtenberg (Eds.), ***Patents, Citations and Innovations: A Window on the Knowledge Economy***: 379-401. Cambridge, MA: MIT Press.

Jansen, J. J. P., Van Den Bosch, F. A. J., & Volberda, H. W. 2006. Exploratory innovation, exploitative innovation, and performance: Effects of organizational antecedents and environmental moderators. ***Management Science***, 52(11): 1661–1674.

Jensen, R.J. & Szulanski, G. 2007. Template use and the effectiveness of knowledge transfer. ***Management Science***, 53: 1716-1730.

Kale, P., Singh, H., & Perlmutter, H. 2000. Learning and protection of proprietary assets in strategic alliances: Building relational capital. ***Strategic Management Journal***, 21(3): 217-238.

Katila, R. & Ahuja, G. 2002. Something old, something new: A longitudinal study of search behavior and new product introduction. ***Academy of Management Journal***, 45: 1183-1194.

Kim, D.J. & Kogut, B. 1996. Technological platforms and diversification. ***Organization Science***, 7: 283-301.

Kogut, B. 1988. Joint ventures: Theoretical and empirical perspectives. ***Strategic Management Journal***, 9: 332.

Kogut, B. 2000. The network as knowledge. ***Strategic Management Journal***, 21: 405-425.

Kogut, B. & Zander, U. 1996. What firms do? Coordination, identity, and learning. ***Organization Science***, 7: 502-518.

Kotabe, M., & Swan, K.S. 1995. The role of strategic alliances in high-technology new product development. ***Strategic Management Journal***, 16: 621-636.

Kreps, D. 1990. Corporate culture and economic theory. In J. Alt and K. Shepsie (Eds.), ***Perspectives on positive political economy*** (pp. 90-143). New York: Cambridge University Press.

Lane, P.J. & Lubatkin, M. 1998. Relative absorptive capacity and interorganizational learning. ***Strategic Management Journal***, 19: 461-477.

Larson, A. 1992. Network dyads in entrepreneurial settings: A study of the governance of exchange relationships. ***Administrative Science Quarterly***, 37: 76-104.

- Laumann, E.O., Marsden, P.V., & Prensky, D. 1983. The boundary specification problem in network analysis. In R.S. Burt & M.J. Minor (Eds.), ***Applied Network Analysis***, 18-34. Beverly Hills: Sage.
- Lavie, D. 2006. The competitive advantage of interconnected firms: An extension of the resource-based view. ***Academy of Management Review***, 21: 825-860.
- Layton, E. 1974. Technology as knowledge. ***Technology and Culture***, 15: 31-41.
- Levin, R.C., Klevorick, A.K., Nelson, R.H. & Winter, S. 1987. Appropriating the returns from industrial research and development. ***Brookings Papers on Economic Activity***, 3: 783-831.
- Levinthal, D.A. & Fichman, M. 1988. Dynamics of interorganizational attachments: Auditor-client relationships. ***Administrative Science Quarterly***, 33: 345-369.
- Levinthal, D. & March, J.G. 1981. A model of adaptive organizational search. ***Journal of Economic Behavior and Organization***, 2: 307-333.
- Liang, K-Y. & Zenger, S. 1986. Longitudinal data analysis using generalized linear models. ***Biometrika***, 73(1): 13-22.
- Luo, Y. 2005. Opportunism in cooperative alliances: Conditions and solutions. In O. Shenkar and J.J. Reuer (Eds.), ***Handbook of Strategic Alliances***, Chapter 4. Thousand Oaks, CA: Sage.
- Lyles, M.A. & Salk, J.E. 1996. Knowledge acquisition from foreign parents in international joint ventures: An empirical examination in the Hungarian context. ***Journal of International Business Studies***, 27: 877-903.
- Macaulay, S. 1963. Non-contractual relations in business: A preliminary study. ***American Sociological Review***, 28: 55-67.
- March, J.G. 1991. Exploration and exploitation in organizational learning. ***Organization Science***, 2: 71-87.
- Martin, X. & Mitchell, W. 1998. The influence of local search and performance heuristics on new design introduction in a new product market. ***Research Policy***, 26: 753-771.
- Maurer, I. & Ebers, M. 2006. Dynamics of social capital and their performance implications: Lessons from biotechnology start-ups. ***Administrative Science Quarterly***, 51: 262-292.
- McEvily, B. & Zaheer, A. 1999. Bridging ties: A source of firm heterogeneity in competitive capabilities. ***Strategic Management Journal***, 20: 1133-1156.
- McGrath, R.G., A. Nerkar. 2004. Real options reasoning and a new look at the R&D investment strategies of pharmaceutical firms. ***Strategic Management Journal***, 25: 1-21.

Mezias, S.J. & Glynn, M.A. 1993. The three faces of corporate renewal: Institution, revolution, and evolution. *Strategic Management Journal*, 14: 77-101.

Mowery, D.C., Oxley, J.E., & Silverman, B.S. 1996. Strategic alliances and interfirm knowledge transfer. *Strategic Management Journal*, 17 (winter special issue): 77-91.

Nahapiet, J. & Ghoshal, S. 1998. Social capital, intellectual capital, and the organizational advantage. *Academy of Management Review*, 23: 242-266.

Narin, F., Noma, E., & Perry, R. 1987. Patents as indicators of corporate technological strength. *Research Policy*, 16: 143-155.

Nelson, R.R. & Winter, S. 1982. *An Evolutionary Theory of Economic Change*. Cambridge, MA: Harvard University Press.

Nerkar, A. 2003. Old is gold? The value of temporal exploration in the creation of new knowledge. *Management Science*, 49: 211-229.

Nonaka, I. 1994. A dynamic theory of organizational knowledge creation. *Organization Science*, 5: 14-37.

Noteboom, B. 1999. Innovation and inter-firm linkages: new implications for policy. *Research Policy*, 28: 793-805.

Obstfeld, D. 2005. Social networks, the tertius lungens orientation, and involvement in innovation. *Administrative Science Quarterly*, 50: 100-130.

Papke, L.W. & Wooldridge, J.M. 2005. Panel data methods for fractional response variables with an application to test pass rates. Working Paper.

Parise, S. & Casher, A. 2003. Alliance portfolios: Designing and managing your network of business-partner relationships. *Academy of Management Executive*, 17(4): 25-39.

Parkhe, A. 1993. Strategic alliance structuring: A game theoretic and transaction cost examination of interfirm cooperation. *Academy of Management Journal*, 36: 794-829.

Patel, P. & Pavitt, K. 1997. The technological competencies of the world's largest firms: complex and path-dependent, but not much variety. *Research Policy*, 26: 141-156.

Pavitt, K. 1988. Uses and abuses of patent statistics. In A.F.J. van Raan (Ed.), *Handbook of Quantitative Studies of Science and Technology* (pp. 509-536). Amsterdam: Elsevier.

Phene, A., Fladmoe-Lindquist, K. & Marsh, L. 2006. Breakthrough innovations in the U.S. biotechnology industry: The effects of technological space and geographic origin. *Strategic Management Journal*, 27: 369-388.

- Pisano, G.P. Using equity participation to support exchange: Evidence from the biotechnology industry. *Journal of Law, Economics and Organization*, 5: 109-126.
- Portes, A. 1998. Social capital: Its origins and applications in modern sociology. *Annual Review of Sociology*, 24: 1-24.
- Quintana-Garcia, C. & Benavides-Velasco, C.A. 2008. Innovative competence, exploration and exploitation: The influence of technological diversification. *Research Policy*, 37: 492-507.
- Raubb, W. & Weesie, J. 1990. Reputation and efficiency in social interactions: An example of network effects. *American Journal of Sociology*, 96: 626-654.
- Reagans, R.W. & McEvily, B. 2003. Network structure and knowledge transfer. *Administrative Science Quarterly*, 48: 240-267.
- Reagans, R.W. Zuckerman, E.W. & McEvily, B. 2004. Two holes in one? Information and control in the analysis of structural advantage. Working Paper.
- Rivette, K.G. 1993. Corporate patent portfolios. *Technology Forecast*, Price Waterhouse.
- Robinson, D.T. & Stuart, T.E. 2007. Network effects in the governance of strategic alliances. *Journal of Law, Economics, and Organization*. 23:242-273.
- Rodan, S. & Galunic, C. 2004. More than network structure: How knowledge heterogeneity influences managerial performance and innovativeness. *Strategic Management Journal*, 25: 541-562.
- Rooks, G., Raub, W., Selten, R. & Tazelaar, F. 2000. How inter-firm co-operation depends on social embeddedness: A vignette study. *Acta Sociologica*, 43: 123-137.
- Rothaermel, F.T. & Deeds, D.L. 2004. Exploration and exploitation alliances in biotechnology: A system of new product development. *Strategic Management Journal*, 25: 201-221.
- Rosenkopf, L. & Nerkar, A. 2001. Beyond local search: Boundary spanning, exploration and impact in the optical disc industry. *Strategic Management Journal*. 22: 287-306.
- Rosenkopf, L. & Almeida, P. 2003. Overcoming local search through alliances and mobility. *Management Science*, 49:751-766.
- Rowley, T., Behrens, D., & Krackhardt, D. 2000. Redundant governance structures: An analysis of structural and relational embeddedness in the steel and semiconductor industries. *Strategic Management Journal*, 21(3): 369-386.
- Sampson, R.C. 2004. Organizational choice in R&D alliances: Knowledge-based and transaction cost perspectives. *Managerial and Decision Economics*, 25: 421-436.

Sampson, R.C. 2005. Experience effects and collaborative returns in R&D alliances. ***Strategic Management Journal***, 26: 1009-1031.

Sampson, R.C. 2007. R&D alliances and firm performance: The impact of technological diversity and alliance organization on innovation. ***Academy of Management Journal***, 50:

Schildt, H.A., Maula, M.V.J. & Keil, T. 2005. Explorative and exploitative learning from external corporate ventures. ***Entrepreneurship Theory and Practice***, 29: 493-515.

Schilling, M. & Phelps, C. 2007. Interfirm collaboration networks and knowledge creation: The impact of large scale network structure on firm innovation. ***Management Science***, 53: 1113-1126.

Schmookler, J. 1966. ***Invention and Economic Growth***. Cambridge, MA: Harvard University Press.

Schumpeter, J.A. 1934. ***The Theory of Economic Development***. Cambridge, MA: Harvard University Press.

Scott, J. 1991. ***Social Network Analysis: A Handbook***. London: Sage.

Shan, W., Walker, G., & Kogut, B. 1994. Interfirm cooperation and startup innovation in the biotechnology industry. ***Strategic Management Journal***, 15: 387-394.

Shaver, J.M. 1998. Accounting for endogeneity when assessing strategy performance: Does entry mode choice affect FDI survival? ***Management Science***, 44: 571-585.

Sidhu, J.S., Commandeur, H.R. & Volberda, H.W. 2007. The multifaceted nature of exploration and exploitation: Value of supply, demand, and spatial search for innovation. ***Organization Science***, 18: 20-38.

Siggelkow, N. & Rivkin, J.W. 2005. Speed and search: Designing organizations for turbulence and complexity. ***Organization Science***, 16: 101-122.

Simonton, D.K. 1999. ***Origins of Genius***. New York: Oxford University Press.

Singh, J.V. 1986. Performance, slack, and risk taking in organizational decision making. ***Academy of Management Journal***, 29: 562-585.

Smith, W.K. & Tushman, M.L. 2005. Managing strategic contradictions: A top management model for managing innovation streams. ***Organization Science***, 16: 522-536.

Smith-Doerr, L., Owen-Smith, J., Koput, K.W., & Powell, W.W. 1999. Networks and knowledge production: Collaboration and patenting in biotechnology. In R.T.A.J. Leenders & S. Gabbay (Eds.), ***Corporate social capital*** (pp. 331-350). Norwell, MA: Kluwer Academic Publishers.

Smith-Doerr, L. & Powell, W.W. 2005. Network and economic life. In N. Smelser and R. Swedberg (Eds.), ***The Handbook of Economic Sociology***, 2nd edition (pp. 379-402). Princeton, NJ: Russell Sage Foundation/Princeton University Press.

Soh, P-H. 2003. The role of networking alliances in information acquisition and its implication for new product performance. ***Journal of Business Venturing***, 18: 727-744.

Sorensen, J.B. & Stuart, T.E. 2000. Aging, obsolescence, and organizational innovation. ***Administrative Science Quarterly***, 45: 81-112.

Stuart, T.E. 2000. Interorganizational alliances and the performance of firms: A study of growth and innovation rates in a high-technology industry. ***Strategic Management Journal***, 21: 791-811.

Stuart, T. & Podolny, J. 1996. Local search and the evolution of technological capabilities. ***Strategic Management Journal***, 17(summer special issue): 21-38.

Szulanski, G. 1996. Exploring internal stickiness: Impediments to the transfer of best practice within the firm. ***Strategic Management Journal***, 17(winter special issue): 27-43.

Teece, D. 1992. Competition, cooperation, and innovation: Organizational arrangements for regimes of rapid technological progress. ***Journal of Economic Behavior and Organization***, 18: 1-25.

Teece, D., Pisano, G., & Shuen, A. 1997. Dynamic capabilities and strategic management. ***Strategic Management Journal***, 18: 509-533.

Tiwana, A. 2008. Do bridging ties complement strong ties? An empirical examination of alliance ambidexterity. ***Strategic Management Journal***, 29: 251-272.

Trajtenberg, M. 1987. Patents, citations, and innovations: Tracing the links. Working paper no. 2457, National Bureau of Economic Research. Washington, D.C.

Uzzi, B. 1996. The sources and consequences of embeddedness for the economic performance of organizations: The network effect. ***American Sociological Review***, 61: 674-698.

Uzzi, B. 1997. Social structure and competition in interfirm networks: The paradox of embeddedness. ***Administrative Science Quarterly***, 42: 35-67.

von Hayek, F. 1945. The use of knowledge in society. ***American Economic Review***, 35: 519-530.

von Hippel, E. 1988. ***The Sources of Innovation***. New York: Oxford University Press.

Walker, G., Kogut, B., & Shan, W. 1997. Social capital, structural holes and the formation of an industry network. ***Organization Science***, 8: 109-125.

Zander, U. & Kogut, B. 1995. Knowledge and the speed of transfer and imitation of organizational capabilities: An empirical test. ***Organization Science***, 6: 76-92.

Table 1: Primary U.S. Patent Classes Used to Represent Telecommunications Equipment^a

Class No.	Title	Class No.	Title
178	Telegraphy	358	Facsimile and & static presentation processing
179 (discontinued)	Telephony	359	Optics: systems (including communication) and elements
329	Demodulators	367	Communications, electrical: acoustic wave systems and & devices
332	Modulators	370	Multiplex communications
333	Wave transmission lines and networks	375	Pulse or digital communications
334	Tuners	379	Telephonic communications
340	Communications: electrical	381	Electrical audio signal processing systems and & devices
341	Coded data generation or conversion	382	Image analysis
342	Communications: directive radio wave systems & devices	385	Optical waveguides
343	Communications: radio wave antennas	455	Telecommunications
348	Television	725	Interactive video distribution systems

^a Because patents are classified by technological and functional principles, they do not map easily to product-based industrial definitions (Griliches, 1990). To identify the technological domain of telecommunication equipment manufacturers I utilized both Silverman's (1996) concordance method and experts. I used the concordance for communications equipment developed by scholars at Science and Technology Policy Research (SPRU), a unit of Sussex University in the UK, and the concordance developed by the Community of Science Inc., an Internet company that provides collaborative tools and services for research scientists and engineers. I identified the primary classes common to both of these expert-based concordances as a baseline concordance. I then compared this list of classes with a rank-ordered list delineating the degree to which International Patent Classes were associated with US SIC 366 as the industry of manufacture as of 1988. In order to make this comparison I used the USPTO's USPC-IPC concordance. The primary classes listed in the baseline concordance were associated with the highest ranked IPC classes associated with US SIC 366 (except for class 725, which did not exist in the late 1980s). This indicates that the 22 primary classes used in this study to represent communications equipment technology (Table 1) are assigned patents that are most frequently associated with SIC 366.

Table 2: Descriptive Statistics and Correlations

	Variable	Mean	SD	Min.	Max.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	Exploratory Innovation	0.718	0.19	0	1	1																	
2	Network Tech. Diversity	0.223	0.332	0	1.71	0.19	1																
3	Density	0.339	0.33	0	1	0.18	-0.27	1															
4	Network Size	6.198	6.334	1	35	-0.15	0.79	-0.36	1														
5	Duration	3.101	1.87	0	14	0.08	0.14	-0.14	0.19	1													
6	Repeated Ties	0.378	0.464	0	4	0.19	0.49	-0.17	0.41	0.13	1												
7	JV	0.18	0.192	0	1	0.02	0.39	-0.18	0.35	0.26	0.20	1											
8	International	0.297	0.254	0	1	-0.06	0.16	-0.14	0.14	-0.02	0.08	0.28	1										
9	Market Overlap	0.327	0.235	0	1	0.02	0.04	0.08	-0.06	-0.02	-0.09	-0.12	0.14	1									
10	Sales (ln)	6.47	2.66	-2.996	11.336	-0.17	0.61	-0.22	0.54	0.25	0.66	0.39	0.16	-0.08	1								
11	Current	2.315	1.583	.003	23.46	0.13	-0.21	0.09	-0.18	-0.19	-0.26	-0.29	-0.17	0.06	-0.25	1							
12	R&D (ln)	3.694	2.641	-4.71	8.66	0.18	0.62	-0.22	0.60	0.23	0.65	0.38	0.17	-0.08	0.93	-0.21	1						
13	Patent Stock	582.056	1267.739	0	6875	-0.14	0.50	-0.21	0.45	0.21	0.67	0.29	0.15	-0.11	0.81	-0.25	0.85	1					
14	Age	45.394	36.033	2	150	0.13	0.58	-0.18	0.52	0.38	0.37	0.50	0.25	-0.04	0.51	-0.30	0.59	0.52	1				
15	Alliance Experience	44.829	64.832	1	416	0.14	0.67	-0.25	0.71	0.26	0.68	0.32	0.09	-0.1	0.83	-0.26	0.86	0.87	0.59	1			
16	Firm Tech. Diversity	0.763	0.307	0	1	-0.09	0.34	-0.19	0.27	0.32	0.25	0.40	0.11	-0.09	0.32	-0.38	0.32	0.30	0.53	0.34	1		
17	Acquisitions	0.96	1.708	0	16	-0.06	0.34	-0.08	0.40	0.03	0.07	0.24	0.06	0	0.2	-0.11	0.26	0.13	0.38	0.30	0.20	1	
18	North America	0.652	0.477	0	1	-0.10	-0.50	0.14	-0.25	-0.30	-0.39	-0.52	-0.29	-0.11	-0.46	0.29	-0.48	-0.38	-0.61	-0.43	-0.43	-0.19	1
19	Europe	0.191	0.393	0	1	0.04	0.51	-0.10	0.34	0.14	0.06	0.50	0.21	0.12	0.18	-0.18	0.24	0.10	0.55	0.24	0.30	0.40	-0.66

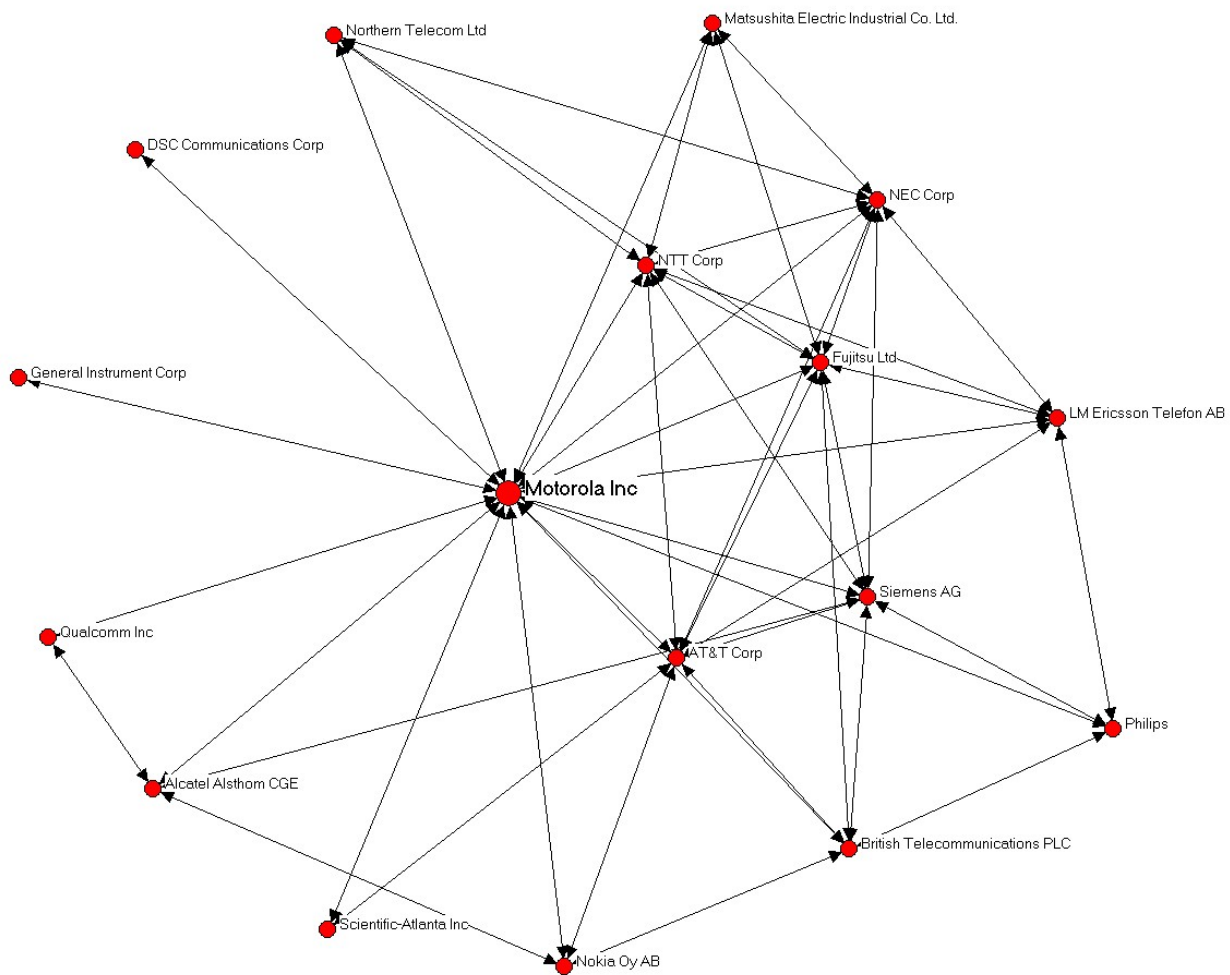
Table 3: Random Effects Panel Regression Predicting Firm Exploratory Innovation

Variable	Model			
	1	2	3	4
Constant	0.609*** (0.069)	0.772*** (0.065)	0.771*** (0.066)	0.769*** (0.066)
Network Size	-0.00249 (0.0018)	-0.00454 (0.00080)	-0.00433 (0.0008)	-0.00400 (0.0009)
Duration	0.000774 (0.0055)	0.00336 (0.0062)	0.00289 (0.0062)	0.00263 (0.0063)
Repeated Ties	0.00343 (0.018)	0.00611 (0.017)	0.00615 (0.017)	0.00693 (0.017)
JV	0.0227 (0.063)	0.0689 (0.059)	0.0631 (0.059)	0.0565 (0.061)
International	-0.0401 (0.036)	-0.0923** (0.037)	-0.0885** (0.038)	-0.0851** (0.038)
Market Overlap	-0.00200 (0.035)	-0.0381 (0.039)	-0.03 (0.039)	-0.0276 (0.039)
Sales (ln)	-0.0225** (0.010)	-0.0151 (0.0096)	-0.0160* (0.0096)	-0.0164* (0.0097)
Current	0.0135** (0.0055)	0.01492** (0.0063)	0.01512** (0.0063)	0.01490** (0.0064)
R&D (ln)	0.0143 (0.0095)	0.0170* (0.0089)	0.0171* (0.0089)	0.0171* (0.0090)
Patent Stock	-0.00000751*** (0.0000012)	-0.0000123*** (0.0000099)	-0.0000118*** (0.0000100)	-0.0000113*** (0.000010)
Age	0.00011 (0.00038)	0.000333 (0.00028)	0.000321 (0.00029)	0.000304 (0.00029)
Alliance Experience	0.000244 (0.00029)	0.000307 (0.00024)	0.000296 (0.00024)	0.000268 (0.00025)
Firm Technological Diversity	0.0122 (0.038)	0.0501 (0.037)	0.051 (0.037)	0.0523 (0.037)
Acquisitions	0.00214 (0.0038)	0.00134 (0.0033)	0.00116 (0.0033)	0.00121 (0.0033)
North America	0.0255* (0.015)	-0.00222* (0.013)	-0.00360* (0.020)	-0.00174 (0.012)
Europe	0.0204 (0.034)	0.000339 (0.026)	0.000314 (0.026)	0.00190 (0.027)
Network Technological Diversity (centered)		0.0552** (0.027)	0.0556** (0.028)	0.0631** (0.031)
Density (centered)			0.0490** (0.022)	0.0601* (0.032)
Net Tech Diversity X Density (centered)				0.0965** (0.48)
Year Dummies Included	Yes	Yes	Yes	Yes
Wald chi-squared test (d.f.)		4.66** (1)	6.35** (2)	8.62** (3)
N; Observations	77; 707	77; 707	77; 707	77; 707

Robust standard errors are in parentheses

*p < 0.1; **p < .05; ***p < .01 (two-tailed tests for all variables)

Figure 1: Motorola's 1992 Ego Network Structure of Technology Alliances



Density = 26.67%
Efficiency = 0.75
Constraint = 0.147

Appendix: Alternative Explanations and Robustness Checks

I considered several alternative explanations and assessed the robustness of the results. I removed the time invariant variables and implemented firm fixed-effects. The results were similar to those obtained using random effects, which is consistent with the insignificant Hausman tests (1978) mentioned earlier. Next, I considered the potential endogeneity of network structure and network diversity. The formation and dissolution of alliances reflect choices made by firms. These choices may be based on expectations of the exploration-enhancing benefits of alliances. This introduces the possibility of a self-selection bias. If the underlying causes for these expectations are unobserved, then the observed effects of alliance-related variables may be influenced by omitted variables, resulting in biased and inconsistent parameter estimates (Shaver, 1998). Network structure may be exogenous for a few reasons. Firms form technology alliances for reasons other than exploratory innovation (Hagedoorn, 1993) and do not easily or quickly alter their alliances to optimize their networks for particular objectives (Maurer & Ebers, 2006). Thus, at any point in time, observed alliance networks are not necessarily structured to maximize firm exploratory innovation and are, at least weakly, exogenous. Last, the configuration of a firm's alliance network is beyond the sole control and influence of any one firm in the network and is therefore not a firm choice variable. Although network diversity may change slowly because of inertia in a firm's alliance relationships and thus may not be optimized for exploratory innovation at a given point in time, the level of diversity in a firm's ego network is largely under its control.

Because endogeneity is an empirical question, I tested for the presence of deleterious endogeneity related to both network density and network diversity. I used Davidson and MacKinnon's test (1993), as implemented by the *dmexogxt* procedure in Stata 10. This test compares the estimated coefficient for the assumed endogenous regressor (e.g., Density or Network Diversity) from OLS fixed effects regression with the estimate obtained from a two-stage instrumental variables fixed effects regression. The null hypothesis is that OLS fixed effects yields a consistent parameter estimate. This procedure requires a valid instrumental variable for the two-stage estimator so the second stage estimates are identified. I used firm technological diversity to instrument for network density and network diversity in separate regressions because it was not significantly correlated with exploratory innovation, but was correlated with density and network diversity. Neither the endogeneity test associated with network density nor network diversity was significant. Thus, the parameter estimates for these variables in Table 3 are not unduly influenced by endogeneity.

I performed additional unreported analyses to assess the robustness of my findings. First, I tested for the presence of a non-linear effect of network diversity by including the square of this measure in the models. In all estimations, (including those mentioned below), the squared term was not statistically significant. Second, I experimented with alternative specifications by removing insignificant variables and then removing all control variables. The results related to the three explanatory variables were robust to these alternative specifications. Third, I estimated the full model using a generalized estimating equation (GEE) approach in which I specified a probit link function and an exchangeable working correlation matrix and computed robust standard errors (Papke & Wooldridge, 2005). Results from this analysis for the three explanatory variables were consistent with those reported in Table 3. Fourth, I substituted Burt's (1992) measures of network Efficiency and Constraint for the density measure discussed above. The results obtained using these alternative measures of ego network closure were statistically stronger, but otherwise consistent with those reported in Table 3. Finally, I used the alternative measure of exploratory innovation discussed above. Because this variable is a count and takes on only non-negative integer values, I estimated the full model with negative binomial panel regression, using year dummies and firm random effects (Hausman, Hall & Griliches, 1984). The results were consistent with those reported in Table 3. Overall, the results of the various robustness analyses provide added support for both hypotheses.